

BEE 208 — Question 11: Stop Motor Test When Temperature Is Unsafe

Control Statement: while loop | Faculty of Engineering, ATU

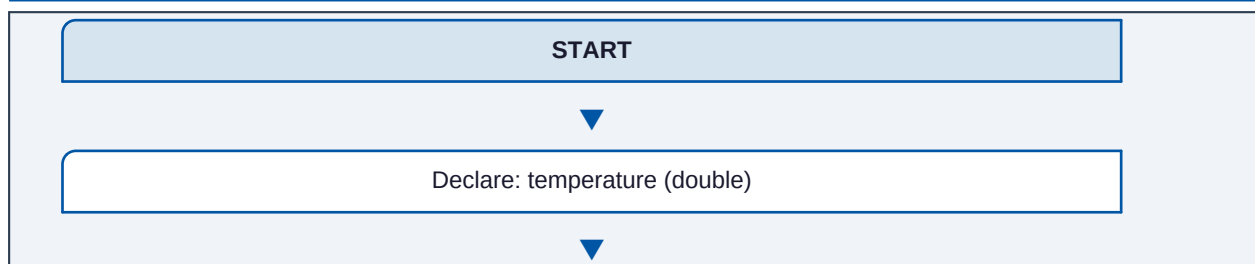
1. ALGORITHM

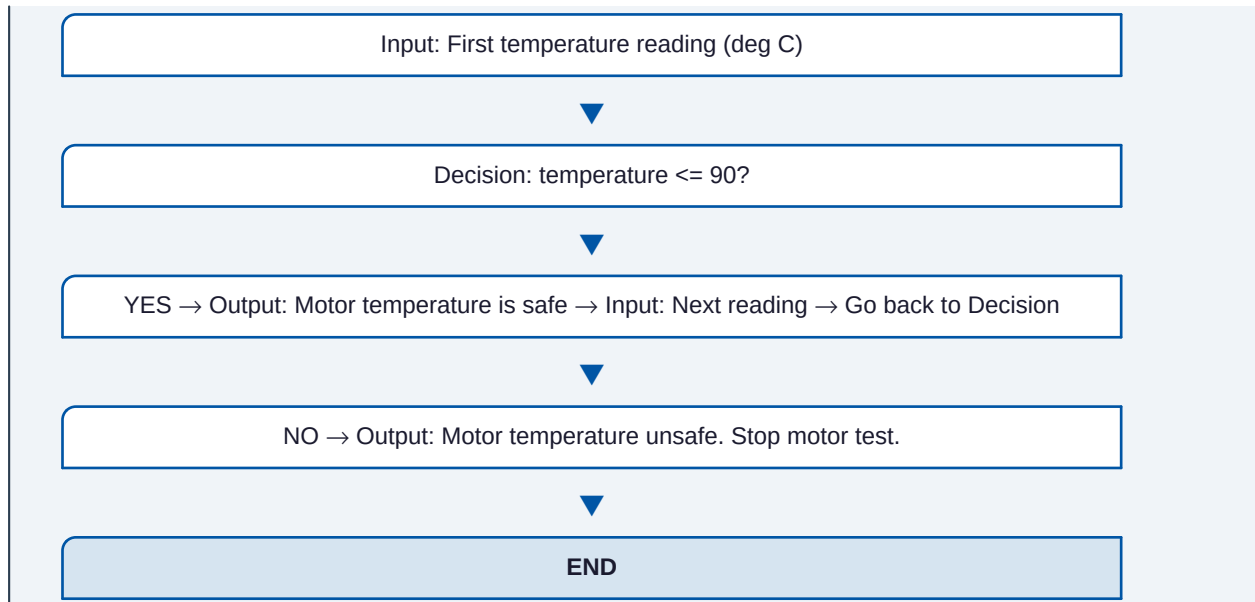
- Step 1: Start
- Step 2: Declare variable temperature (double)
- Step 3: Prompt user to enter first temperature reading
- Step 4: Read temperature
- Step 5: While temperature ≤ 90 , display 'Motor temperature is safe'
- Step 6: Prompt and read next temperature reading
- Step 7: When temperature > 90 , exit loop
- Step 8: Display 'Motor temperature unsafe. Stop motor test.'
- Step 9: End

2. PSEUDOCODE

```
BEGIN  
  
DECLARE temperature AS DOUBLE  
  
OUTPUT 'Enter first temperature reading (deg C):'  
  
INPUT temperature  
  
WHILE temperature  $\leq 90$  DO  
    OUTPUT 'Motor temperature is safe:', temperature, 'deg C'  
    OUTPUT 'Enter next temperature reading (deg C):'  
    INPUT temperature  
END WHILE  
  
OUTPUT 'Motor temperature unsafe. Stop motor test.'  
  
END
```

3. FLOWCHART (Structured Diagram)





4. TEST RESULTS

Test Item	Value / Description	Test Item	Value / Description
Input	85, 88, 91	Expected Output	Safe×2, then: Motor temperature unsafe. Stop motor test.
Input	95 (first entry)	Expected Output	Immediately: Motor temperature unsafe. Stop motor test.